DATA ANALYST TECHNICAL ASSESSMENT

DELIVERY - ON THE GO | OPERATIONS CAPACITY & COST REVIEW

Operational analysis: 13-26 October 2026
Forecast horizon: 27 October-23 November 2026

Core decision: Reject the blanket request for eight new agents. Recommend two targeted hires, two same-cluster transfers and a Riverside hold-reduction programme.

22,789 orders	25,595 assigned attempts	21,078 delivered outcomes	3 / 6 hubs above target in Week 2	BDT 62,800 recommended first-month spend

Submission status

This package contains the analysis, editable PostgreSQL SQL, Streamlit source, results, dashboard UI previews and an eight-slide presentation. The Streamlit app has not been publicly deployed. A live public URL must be added after deployment; the current PDF deliberately does not invent one.

Dashboard URL: Pending public deployment - see deployment/deployment_guide.md

Executive summary

The second week shows a 16.34% increase in new orders and an 18.97% increase in assigned attempts. Capacity is uneven: H01, H03 and H04 exceed the 85% planning target, while H02 and H05 have under-used capacity. H03 is the outlier on process quality: 64.42% of its Week 2 holds are agent-side, and Late Dispatch is the largest reason.

Theme	Finding
Volume	Orders increased from 10,534 to 12,255 (+16.34%); attempts rose from 11,689 to 13,906 (+18.97%).
Capacity	Week 2 utilisation exceeds 85% at H01 (88.64%), H03 (90.80%) and H04 (95.87%).
Process	H03 logged 874 Week 2 holds; 563 (64.42%) were agent-side and 312 were Late Dispatch.
Returns	Three H05 agents pass Q5's tenure and >15% return-ratio criteria.
Staffing	Q7's gross peak shortfall is 5 agents across H01, H03 and H04 before transfer/intervention.
Recommendation	Transfer 2 H02 Fixed agents to H01, hire 2 Fixed agents at H04, and fund the H03 hold-reduction programme.

Financial recommendation

The recommended plan costs BDT 62,800 in the first month and leaves BDT 7,200 from the BDT 70,000 budget. A linear 14-day hold-programme ramp is modelled to prevent about 856 H03 agent-side holds and save about BDT 25,682 in hold charges over four weeks. This is a scenario estimate, not guaranteed cash savings.

Decision caveat

Riverside remains above the 85% planning target during the first two programme-ramp weeks (87.9% and 86.1% of standard capacity) but below the 100% standard workload. Management must monitor attempts per agent daily and use the BDT 7,200 reserve only if a measured trigger is breached.

Data validation and methodology

All eight supplied CSV tables were loaded. Primary keys are unique, joins across orders, agents and hubs resolve, and every Assigned attempt has exactly one Delivered, Hold or Returned outcome. The final status in orders matches the last logged outcome for every order. The 48,148 null values in order_logs.hold_reason are expected on non-Hold events; every Hold event has a reason.

Source table	Rows
hubs.csv	6
agents.csv	64
orders.csv	22789
order_logs.csv	51190
agent_salary.csv	2
costs.csv	5
rules.csv	8
volume_forecast.csv	24

Definitions used throughout

Date windows: Week 1 = 13-19 October 2026; Week 2 = 20-26 October 2026. The forecast covers 27 October-23 November 2026.

Attempt: Every Assigned event is one attempt. Every attempt is matched to exactly one outcome event.

Utilisation: Assigned attempts / (available agents × 40 attempts/day × 7 calendar days). Week 1 includes all agents; Week 2 excludes the two Inactive agents who left at the end of Week 1.

Agent-side hold: Late Dispatch, Not Attempted and Agent Time Shortage. All other hold reasons are customer-side.

Q4 held order: A unique order with at least one Hold event; its outcome is taken from orders.final_status.

Q5 return ratio: Returned outcome events / Assigned attempts per agent; tenure measured on 26 October and strictly greater than 30 days.

Q6 working day: An agent-day with at least one Assigned event. Fixed = monthly salary / 26 per worked day. Variable = daily allowance per worked day + allowance per Delivered outcome. Hold/return charges accrue in the week of the event.

Q7 forecast: Expected orders × 14-day observed attempts-per-order ratio by hub; required agents = CEIL(forecast attempts / (7 × 40 × 85%)).

Strategic scenario assumptions

For Q8/Q9 Riverside stress testing, the more recent Week 2 attempts-per-order rate is used. The hold programme reaches a 50% reduction of agent-side holds after 14 days; because the brief does not state the intermediate curve, a linear ramp from 27 October is assumed. For new hires, the first seven days are modelled at 60% of 40 attempts/day (24/day), then 85% of standard (34/day) for the remainder of 26 paid workdays.

Task 1 - SQL assessment

Q1. Daily Agent Productivity

The daily result contains 882 employed agent-days. All available roster days are retained, including zero-assignment days. Inactive agents are included only through 19 October. The table below shows the first 20 rows; the complete 882-row daily file is provided at outputs/query_results/q1_daily_agent_productivity.csv.

Date	Agent ID	Agent name	Hub	Assigned	Delivered	Hold	Returned
2026-10-13	A1001	Weno Tremaine	H01	20	18	2	0
2026-10-13	A1002	Valel Wexley	H01	21	17	4	0
2026-10-13	A1003	Torian Vantell	H01	27	22	3	2
2026-10-13	A1004	Garian Larkin	H01	20	19	1	0
2026-10-13	A1005	Wenan Brannick	H01	23	20	2	1
2026-10-13	A1006	Toren Rowntree	H01	26	23	2	1
2026-10-13	A1007	Coris Rowntree	H01	27	25	0	2
2026-10-13	A1008	Garon Holloway	H01	26	22	3	1
2026-10-13	A1009	Perian Halvorsen	H01	34	27	6	1
2026-10-13	A1010	Joris Fenwick	H01	28	23	2	3
2026-10-13	A1011	Marel Fenwick	H01	14	12	1	1
2026-10-13	A1012	Zarik Rowntree	H02	19	18	1	0
2026-10-13	A1013	Pero Ketteridge	H02	25	22	2	1
2026-10-13	A1014	Zaris Quorne	H02	26	23	2	1
2026-10-13	A1015	Toren Larkin	H02	24	21	3	0
2026-10-13	A1016	Noren Vantell	H02	17	15	0	2
2026-10-13	A1017	Joran Pelham	H02	23	23	0	0
2026-10-13	A1018	Lorus Brannick	H02	14	13	1	0
2026-10-13	A1019	Selus Rowntree	H02	19	17	2	0
2026-10-13	A1020	Noris Fenwick	H02	15	14	1	0

Full hub summary

Hub	Name	Cluster	Delivered	Agent-days	Avg deliveries / agent / day
H01	Greenfield	Cluster A	3908	147	26.59

Hub	Name	Cluster	Delivered	Agent-days	Avg deliveries / agent / day
H02	Lakeview	Cluster A	3001	154	19.49
H03	Riverside	Cluster B	3479	168	20.71
H04	Hillcrest	Cluster B	3798	126	30.14
H05	Brookside	Cluster C	3411	154	22.15
H06	Westwood	Cluster C	3481	133	26.17

Q2. Hub Utilisation in Week 1 and Week 2

Utilisation is defined as Assigned attempts / (available agents × 40 attempts per day × 7 days). The Week 2 flag compares utilisation with the 85% planning target.

Hub	Name	W1 attempts	W2 attempts	W1 agents	W2 agents	W1 util.	W2 util.	Above 85%?	Attempt change
H01	Greenfield	1925	2482	11	10	62.50%	88.64%	Yes	+28.94%
H02	Lakeview	1647	1730	11	11	53.47%	56.17%	No	+5.04%
H03	Riverside	2281	3051	12	12	67.89%	90.80%	Yes	+33.76%
H04	Hillcrest	1943	2416	9	9	77.10%	95.87%	Yes	+24.34%
H05	Brookside	2023	2202	11	11	65.68%	71.49%	No	+8.85%
H06	Westwood	1870	2025	10	9	66.79%	80.36%	No	+8.29%

Week 2 above target: H01 Greenfield (88.64%), H03 Riverside (90.80%) and H04 Hillcrest (95.87%).

Q3. Hold Reasons

Week 2 hold events total 1,679. Agent-side holds total 740 (44.07%) across the six hubs. Riverside alone accounts for 563 agent-side holds.

Hub	Name	Holds	Agent-side	Agent share	Top reason	Count	Second reason	Count
H01	Greenfield	168	32	19.05%	Customer Unreachable	73	Customer Rescheduled	35
H02	Lakeview	123	26	21.14%	Customer Unreachable	50	Customer Rescheduled	25
H03	Riverside	874	563	64.42%	Late Dispatch	312	Customer Unreachable	156
H04	Hillcrest	182	40	21.98%	Customer Unreachable	77	Customer Rescheduled	39
H05	Brookside	186	49	26.34%	Customer Unreachable	68	Customer Rescheduled	43
H06	Westwood	146	30	20.55%	Customer Unreachable	66	Customer Rescheduled	28

The agent-side classification follows the exact three reasons in the case notes: Late Dispatch, Not Attempted and Agent Time Shortage. Riverside's top reason is Late Dispatch (312).

Q4. What Happens to Held Orders

The query identifies unique orders that had at least one Hold event during the 14-day period, then classifies them by final_status. Of 2,599 held orders, 2,118 (81.49%) were finally delivered, 245 (9.43%) returned and 236 (9.08%) remain on hold.

Hub	Name	Held orders	Delivered	Deliv. share	Returned	Return share	Still hold	Hold share
H01	Greenfield	303	269	88.78%	8	2.64%	26	8.58%
H02	Lakeview	228	204	89.47%	10	4.39%	14	6.14%
H03	Riverside	1184	903	76.27%	165	13.94%	116	9.80%
H04	Hillcrest	316	272	86.08%	15	4.75%	29	9.18%
H05	Brookside	324	260	80.25%	37	11.42%	27	8.33%
H06	Westwood	244	210	86.07%	10	4.10%	24	9.84%

Q5. High-Return Agents

Return ratio is Returned outcome events divided by Assigned attempts. Only agents with more than 30 days of tenure on 26 October 2026 and a return ratio above 15% are included.

Agent ID	Agent name	Hub	Salary type	Tenure (days)	Assigned	Returned	Return ratio
A1045	Belis Rowntree	H05	Variable	463	349	94	26.93%
A1046	Coris Halvorsen	H05	Fixed	647	414	91	21.98%
A1047	Wenian Vantell	H05	Variable	526	413	72	17.43%

All three qualifying agents are in H05 Brookside: A1045 (26.93%), A1046 (21.98%) and A1047 (17.43%). These are process-review flags, not proof of individual fault; investigate route, customer and hand-off context.

Q6. Cost per Delivered Order

Costs are allocated to the week in which the operational events occur. A workday is inferred when an agent has at least one Assigned event on that date. Variable per-delivery allowance is charged on Delivered outcomes for Variable agents. Delivered outcome events in the same week form the denominator. COD amount and customer delivery charge are excluded because they are not listed as operating costs in the cost sheet.

Hub	Week	Fixed pay	Var daily	Var / delivery	Hold cost	Return cost	Total cost	Delivered	BDT / delivered
H01	Week 1	31,200	8,100	8,358	4,770	3,740	56,168	1,698	33.08
H01	Week 2	27,300	8,100	12,544	5,040	5,720	58,704	2,210	26.56
H02	Week 1	35,100	6,000	5,628	3,630	3,135	53,493	1,469	36.41
H02	Week 2	35,100	6,300	6,188	3,690	4,125	55,403	1,532	36.16
H03	Week 1	36,400	8,100	6,538	18,810	8,580	78,428	1,498	52.36
H03	Week 2	35,100	8,400	9,982	26,220	10,780	90,482	1,981	45.67
H04	Week 1	9,100	14,400	18,396	4,830	5,225	51,951	1,687	30.79
H04	Week 2	8,450	14,400	23,226	5,460	6,765	58,301	2,111	27.62
H05	Week 1	16,900	13,800	15,386	5,130	12,155	63,371	1,631	38.85
H05	Week 2	17,550	14,400	15,946	5,580	12,980	66,456	1,780	37.33
H06	Week 1	35,100	3,900	4,592	3,720	3,245	50,557	1,687	29.97
H06	Week 2	29,250	4,200	6,062	4,380	4,675	48,567	1,794	27.07

H03 remains the most expensive hub in both weeks: BDT 52.36 in Week 1 and BDT 45.67 in Week 2. The improvement in Week 2 does not remove its cost gap; it still has the highest hold and return burden.

Q7. Agents Needed for the Forecast

The forecast uses the 14-day observed attempts-per-order rate for each hub. Required agents = CEIL(forecast orders × attempts per order / (7 × 40 × 85%)). A positive staffing gap is a shortage; a negative gap is a surplus.

Hub	Name	Start	End	Forecast orders	Attempts / order	Required	Active	Gap
H01	Greenfield	27 Oct	02 Nov	2470	1.073	12	10	2
H02	Lakeview	27 Oct	02 Nov	1544	1.073	7	11	-4
H03	Riverside	27 Oct	02 Nov	2170	1.351	13	12	1
H04	Hillcrest	27 Oct	02 Nov	2264	1.078	11	9	2
H05	Brookside	27 Oct	02 Nov	1955	1.085	9	11	-2
H06	Westwood	27 Oct	02 Nov	1852	1.067	9	9	0
H01	Greenfield	03 Nov	09 Nov	2545	1.073	12	10	2
H02	Lakeview	03 Nov	09 Nov	1591	1.073	8	11	-3
H03	Riverside	03 Nov	09 Nov	2230	1.351	13	12	1
H04	Hillcrest	03 Nov	09 Nov	2333	1.078	11	9	2
H05	Brookside	03 Nov	09 Nov	2015	1.085	10	11	-1
H06	Westwood	03 Nov	09 Nov	1909	1.067	9	9	0
H01	Greenfield	10 Nov	16 Nov	2570	1.073	12	10	2
H02	Lakeview	10 Nov	16 Nov	1606	1.073	8	11	-3
H03	Riverside	10 Nov	16 Nov	2245	1.351	13	12	1
H04	Hillcrest	10 Nov	16 Nov	2356	1.078	11	9	2
H05	Brookside	10 Nov	16 Nov	2035	1.085	10	11	-1
H06	Westwood	10 Nov	16 Nov	1928	1.067	9	9	0
H01	Greenfield	17 Nov	23 Nov	2495	1.073	12	10	2
H02	Lakeview	17 Nov	23 Nov	1559	1.073	8	11	-3
H03	Riverside	17 Nov	23 Nov	2200	1.351	13	12	1
H04	Hillcrest	17 Nov	23 Nov	2287	1.078	11	9	2
H05	Brookside	17 Nov	23 Nov	1975	1.085	10	11	-1
H06	Westwood	17 Nov	23 Nov	1871	1.067	9	9	0

The peak gross shortage before transfers/intervention is five agents: H01 needs two, H03 one and H04 two. H02 and H05 have surplus capacity; transfer eligibility must still respect the same-cluster rule.

Task 2 - Strategic Analysis

Q8. The Hiring Request

Decision: reject the request for eight additional agents. The forecast supports a targeted response rather than a blanket expansion.

Hub	Active	Peak req.	Recommended action	First-month cost / note
H01 Greenfield	10	12	Receive 2 Fixed transfers from H02	BDT 4,000 transfer fees
H02 Lakeview	11	8	Transfer A1018 and A1015; 9 remain	Donor remains above peak need
H03 Riverside	12	13 base	Fund hold programme; no immediate hire	BDT 15,000 programme
H04 Hillcrest	9	11	Hire 2 Fixed agents	BDT 43,800 incl. salary + hiring
H05 Brookside	11	10	No hire; review 3 high-return agents	No incremental staffing cost
H06 Westwood	9	9	No staffing change	No incremental staffing cost

Transfer candidates A1018 Lorus Brannick (Fixed) and A1015 Toren Larkin (Fixed) have H02 delivery ratios of 92.78% and 90.86%, respectively, with return ratios of 3.09% and 2.65%. Both moves stay within Cluster A. H05 is also under 75% utilisation, but cannot move agents to shortage hubs in other clusters; H06 does not need an extra agent.

Riverside has 563 agent-side holds in Week 2 (64.42% of its 874 holds); Late Dispatch accounts for 312. This calls for process correction rather than automatically adding a permanent headcount.

Q9. The Budget

The recommended plan costs BDT 62,800 and leaves BDT 7,200 from the BDT 70,000 allocation.

Use	Calculation	BDT
Two Fixed salaries at H04	2 × 16,900	33,800
Hiring charges	2 × 5,000	10,000
Transfers H02 → H01	2 × 2,000	4,000
H03 hold-reduction programme	1 × 15,000	15,000
Total		62,800
Remaining budget	70,000 - 62,800	7,200

Fixed is the cheaper H04 salary type at the expected workload. New-agent first-month attempt capacity is modelled as (7 × 24) + (19 × 34) = 814 attempts. H04's historical delivered-to-assigned rate is 87.13%, implying about 709 deliveries. Fixed pay is BDT 16,900; estimated Variable pay is BDT 7,800 daily allowance + about BDT 9,929 delivery allowance = BDT 17,729.

Riverside programme return

Forecast week	Forecast orders	Avoided holds	H03 utilisation	Hold charge saving
27 Oct	2170	70	87.9%	2,092
03 Nov	2230	215	86.1%	6,449
10 Nov	2245	289	84.5%	8,657
17 Nov	2200	283	82.8%	8,484

Across four weeks, the model estimates 856 avoided agent-side holds and BDT 25,682 in avoided BDT 30 hold charges. After the BDT 15,000 programme cost, direct net benefit is BDT 10,682 (about 71.2% net ROI). The programme also frees repeat-attempt capacity; no additional revenue is assumed.

Assumptions: programme starts on 27 October; effect ramps linearly to full 50% reduction after 14 calendar days. The brief states the 14-day end point but not the ramp path. The first two Riverside forecast weeks remain at 87.9% and 86.1% of standard workload - above the 85% planning target, but below the 100% standard.

STRICT 85% ALTERNATIVE

Replace the H03 hold-reduction programme with one Variable hire at Riverside. Keeping two Fixed hires at H04 and two H02-to-H01 transfers, the estimated first-month cost is BDT 68,036 (including BDT 15,236 Variable pay and a BDT 5,000 hiring fee), leaving BDT 1,964. This option meets the 85% planning target earlier, but does not fix the agent-side dispatch problem and gives up the modelled BDT 25,682 of avoided hold charges.

Q10. The Risks

Risk	Leading indicator	First-two-week action / trigger
Riverside programme under-delivers	Agent-side holds per order vs 25.71% Week 2 baseline	Confirm daily intervention actions; if reduction is not visible by Day 14, add temporary capacity or revise process.
Riverside workload above plan	Attempts per agent/day; target 34, standard 40	Plan is 35.2 and 34.4 per agent/day in Weeks 1-2. Escalate if average exceeds 40 or backlog / service levels worsen.
H04 hires ramp slowly	Attempts per new agent/day	Expect 24/day for first seven days. Buddy new hires and escalate if output falls materially below ramp assumption.
H02 has insufficient donor buffer	H02 after transfer = 9 agents; peak requirement = 8	Do not transfer a third agent. Monitor weekly utilisation and actual orders.
H05 returns remain high	A1045 26.93%, A1046 21.98%, A1047 17.43%	Review reason codes, route/hand-off evidence and coaching needs; do not treat return ratio alone as misconduct.
Forecast is understated	Actual orders / attempts per order vs forecast	Proposed trigger: investigate if daily orders exceed plan by 10% or attempts/order rises 5% above modelled rate.

Decision rule: keep the two H04 hires and H02→H01 transfers. Continue the H03 no-hire approach only while workload stays below the 40-attempt standard and the hold programme shows measurable progress. Use the BDT 7,200 reserve only against an observed trigger.

Task 3 - Streamlit Dashboard

The source project contains a multi-page Streamlit app with Executive Overview, Hub Performance, Agent Productivity, Holds & Returns, Forecast & Staffing, and Cost Efficiency pages. These are data-backed UI preview images generated from the same calculated metrics. They are not browser captures from a running Streamlit server: Streamlit was unavailable in the offline build environment. Re-capture these pages after running the app locally or deploying it.

Public dashboard URL: Pending deployment. Do not submit a local localhost URL. Deployment steps are in deployment/deployment_guide.md.

Page 1 - Executive overview

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OPERATIONS INTELLIGENCE

- Executive overview
- Hub performance
- Agent productivity
- Holds & returns
- Forecast & staffing
- Cost efficiency

FILTERS

Period

13 Oct – 26 Oct 2026

Hubs

All 6 hubs

UI PREVIEW · DATA BACKED

OPERATIONS CONTROL TOWER / 2026

Executive overview

Decision-ready view of hub capacity, delivery outcomes and the next four weeks.

ORDERS · 14 DAYS	DELIVERED OUTCOMES	ACTIVE AGENTS	HUBS ABOVE TARGET
22,789	21,078	62	3 / 6
+16.3% in Week 2	92.5% of original orders	2 left after Week 1	H01 · H03 · H04

Daily demand and workload

Assigned attempts vs new orders



Where to act first

Different hubs need different solutions

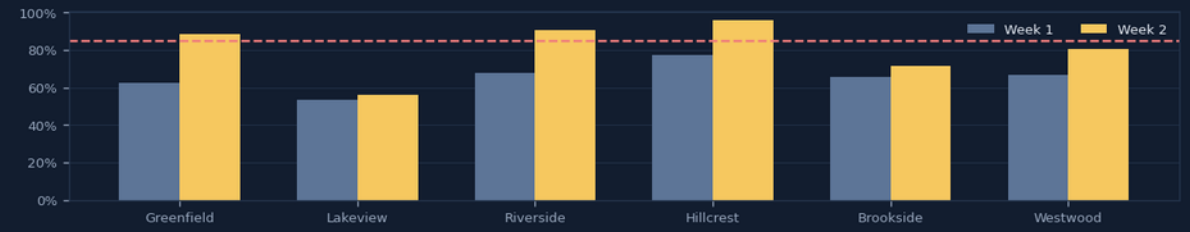
H04 / HIRING
95.87% Week 2 utilisation
Forecast requires 11 agents vs 9 active

H03 / PROCESS
563 agent-side holds
Late Dispatch is the leading hold reason

H01 / TRANSFER
88.64% Week 2 utilisation
H02 is an eligible same-cluster source

Week 2 hub utilisation

Planning target: 85% of the 40-attempt daily standard



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UI PREVIEW · DATA BACKED

OPERATIONS CONTROL TOWER / 2026

Hub performance

Compare workload utilisation, demand growth and forecast requirements by hub.

ABOVE 85% IN WEEK 2

3 hubs

H01 · H03 · H04

FASTEST GROWTH

H03 · +33.8%

Attempt volume vs Week 1

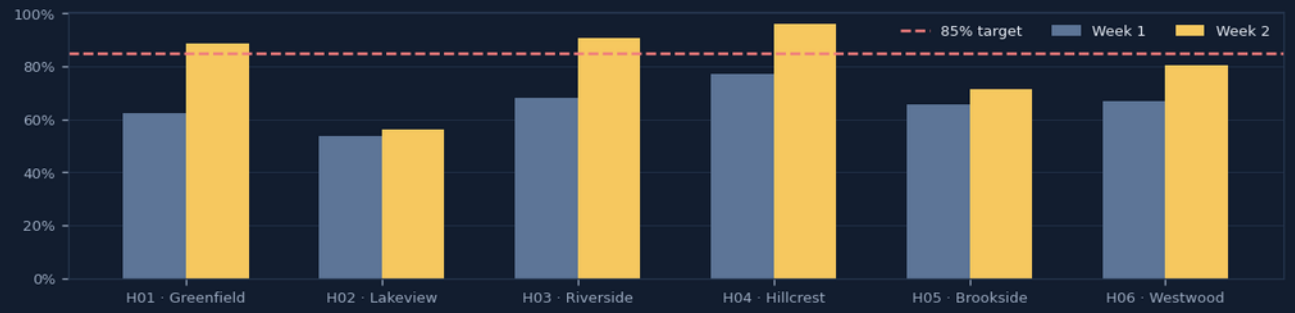
PEAK UTILISATION

H04 · 95.9%

Most constrained

Weekly utilisation by hub

Utilisation = assigned attempts / (available agents × 40 × 7)



Capacity table

Week 1 includes all 64 rostered agents; Week 2 includes 62 Active agents.

Hub	W1 attempts	W2 attempts	W1 agents	W2 agents	W2 utilisation	Above target
H01	1925	2482	11	10	88.6%	YES
H02	1647	1730	11	11	56.2%	NO
H03	2281	3051	12	12	90.8%	YES
H04	1943	2416	9	9	95.9%	YES
H05	2023	2202	11	11	71.5%	NO
H06	1870	2025	10	9	80.4%	NO

DELIVERY / OTG

OPERATIONS INTELLIGENCE

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All 6 hubs

UI PREVIEW · DATA BACKED

OPERATIONS CONTROL TOWER / 2026

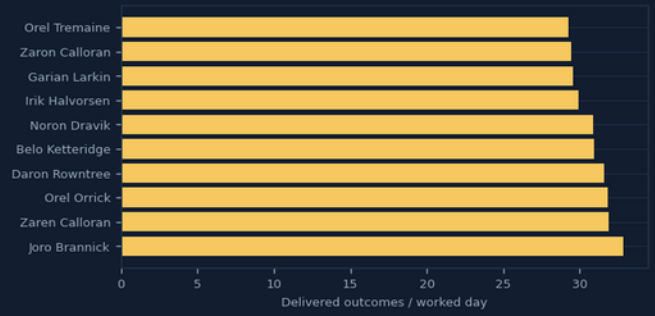
Agent productivity

Daily output, agent-level delivery rates and targeted return-risk review.

AGENTS IN ROSTER	Q5 HIGH-RETURN AGENTS	HIGHEST RETURN RATIO
64	3	26.9%
62 Active after Week 1	All in H05 Brookside	A1045 · H05

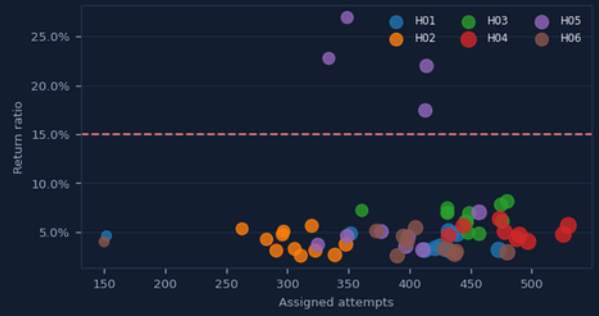
Top agents by deliveries per worked day

Top 10 agents across the 14-day period



Return ratio vs assigned attempts

Dashed line marks the 15% Q5 threshold



Q5 agents requiring review

Tenure >30 days on 26 Oct 2026 and return ratio >15%.

A1045 · Belis Rowntree	H05	Variable	94 / 349 returns / attempts	26.9%
A1046 · Coris Halvorsen	H05	Fixed	91 / 414 returns / attempts	22.0%
A1047 · Wenian Vantell	H05	Variable	72 / 413 returns / attempts	17.4%

DELIVERY / OTG

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- **Holds & returns**
- Forecast & staffing
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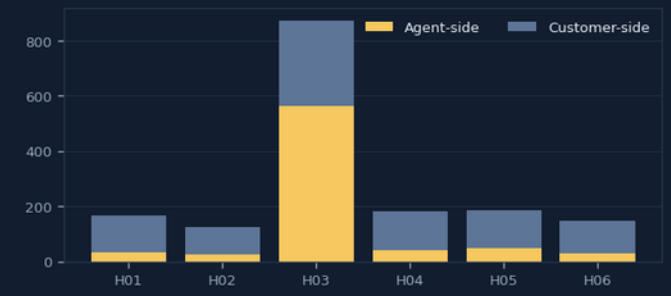
Holds & returns

Separate agent-side failures from customer friction and trace the final outcomes of held orders.

WEEK 2 HOLDS	AGENT-SIDE HOLDS	HELD ORDERS	FINALLY DELIVERED
1,679	740	2,599	2,118
All six hubs	44.1% of Week 2 holds	Held at least once	81.5% of held orders

Week 2 hold mix by hub

Agent-side = Late Dispatch, Not Attempted, Agent Time Shortage



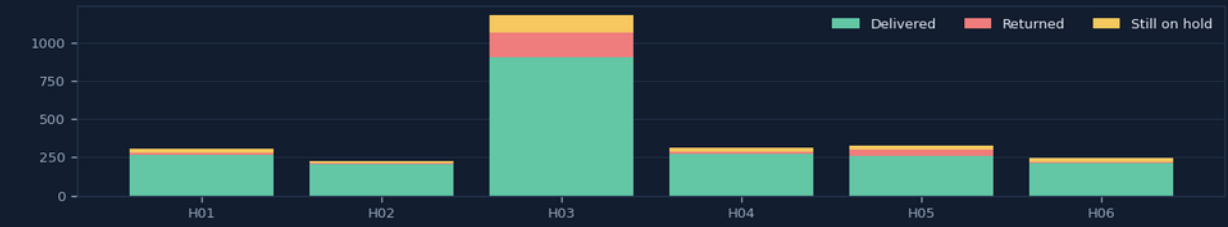
Top reason at each hub

Riverside's dominant reason is Late Dispatch.

H01	Customer Unreachable (73)	2nd: Customer Rescheduled (35)
H02	Customer Unreachable (50)	2nd: Customer Rescheduled (25)
H03	Late Dispatch (312)	2nd: Customer Unreachable (156)
H04	Customer Unreachable (77)	2nd: Customer Rescheduled (39)
H05	Customer Unreachable (68)	2nd: Customer Rescheduled (43)
H06	Customer Unreachable (66)	2nd: Customer Rescheduled (28)

Final outcome of orders held at least once

Unique orders by final order status



DELIVERY / OTG

OPERATIONS INTELLIGENCE

- Executive overview
- Hub performance
- Agent productivity
- Holds & returns
- Forecast & staffing
- Cost efficiency

FILTERS

Period

13 Oct – 26 Oct 2026

Hubs

All 6 hubs

UI PREVIEW · DATA BACKED

OPERATIONS CONTROL TOWER / 2026

Forecast & staffing

Convert forecast orders into workload and show a hub-level response constrained by policy and budget.

FORECAST HORIZON	PEAK GROSS GAP	RECOMMENDED HIRES	BUDGET SPEND
4 weeks	5 agents	2 Fixed	BDT 62,800
27 Oct – 23 Nov	Before transfers / intervention	Both at H04	BDT 7,200 reserve

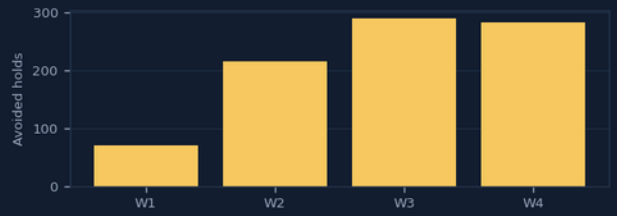
Required agents by forecast week

14-day attempts-per-order rate, 85% planning target



Riverside · hold programme

50% maximum reduction reached after 14 days



Hub-by-hub action

No blanket approval of the eight-agent request

H01	Transfer +2 from H02
H02	Transfer -2; 9 remain
H03	Hold programme; no hire
H04	Hire +2 Fixed agents
H05	No hire; retain surplus
H06	No change

DELIVERY / OTG

OPERATIONS INTELLIGENCE

- Executive overview
- Hub performance
- Agent productivity
- Holds & returns
- Forecast & staffing
- **Cost efficiency**

FILTERS

Period

13 Oct – 26 Oct 2026

Hubs

All 6 hubs

UI PREVIEW · DATA BACKED

OPERATIONS CONTROL TOWER / 2026

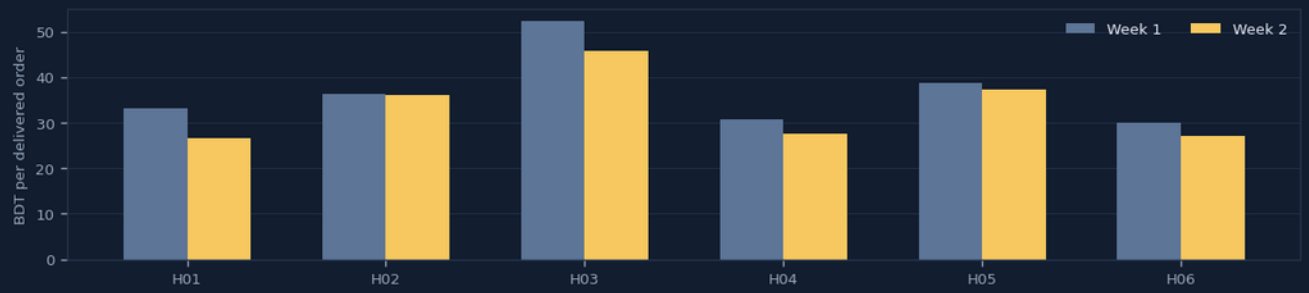
Cost efficiency

See what each successful delivery costs and where hold/return charges are concentrated.

LOWEST W2 COST / DELIVERY	HIGHEST W2 COST / DELIVERY	LARGEST W2 HOLD CHARGE
BDT 26.56	BDT 45.67	BDT 26,220
H01 Greenfield	H03 Riverside	H03 Riverside

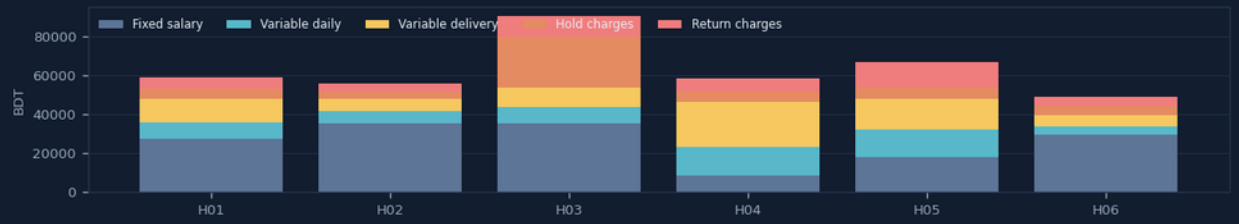
Cost per delivered order

Costs are assigned to the week in which events occur.



Week 2 operating cost mix

Hold and return charges are a material issue at Riverside and Brookside.



Task 4 - Presentation Slides

An eight-slide PowerPoint deck and matching PDF are included separately in report/. The deck is designed for the Head of Operations and covers the executive decision, operational performance, Riverside root cause, staffing forecast, targeted staffing plan, BDT 70,000 budget and two-week risk controls.

Before submission

1. Deploy the Streamlit app publicly and verify it opens in an incognito browser without login.
2. Replace the dashboard URL placeholder in this PDF with the live URL and refresh the dashboard screenshots from the deployed app.
3. Ensure the submission email matches the application email, then upload the updated CV, final PDF, PPTX and dashboard link in the assessment form.

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```
Delivery - On The Go | Data Analyst Technical Assessment
SQL dialect: PostgreSQL

Important interpretation notes:
1) One Assigned event is one attempt; each attempt has one outcome event.
2) Dates in the task are inclusive: Week 1 = 2026-10-13 to 2026-10-19;
   Week 2 = 2026-10-20 to 2026-10-26.
3) Week 1 headcount includes all agents. Inactive agents left at the end of Week 1,
   so Week 2 headcount includes only Active agents.
4) Utilisation = Assigned attempts / (available agents * 40 * 7).
5) Q6 cost allocation is on the event week. A workday is inferred from at least one
   Assigned event for that agent on that date.
*/

-- ============================================================
-- Q1. Daily Agent Productivity
-- A complete roster-date grid is retained; no-assignment days show zeroes.
-- The first 20 rows are obtained with ORDER BY ... LIMIT 20.
-- ============================================================

WITH dates AS (
    SELECT d::date AS work_date
    FROM generate_series(
        DATE '2026-10-13',
        DATE '2026-10-26',
        INTERVAL '1 day'
    ) AS s(d)
),
agent_days AS (
    SELECT
        a.agent_id,
        a.agent_name,
        a.hub_id,
        a.salary_type,
        a.joining_date,
        a.active_status,
        d.work_date
    FROM agents a
    CROSS JOIN dates d
    WHERE a.active_status = 'Active'
       OR (a.active_status = 'Inactive' AND d.work_date <= DATE '2026-10-19')
),
daily_events AS (
    SELECT
        o.hub_id,
        l.agent_id,
        l.event_time::date AS work_date,
        COUNT(*) FILTER (WHERE l.status = 'Assigned') AS assigned_attempts,
        COUNT(*) FILTER (WHERE l.status = 'Delivered') AS delivered_attempts,
        COUNT(*) FILTER (WHERE l.status = 'Hold') AS hold_attempts,
        COUNT(*) FILTER (WHERE l.status = 'Returned') AS returned_attempts
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    WHERE l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY o.hub_id, l.agent_id, l.event_time::date
),
q1_daily AS (
    SELECT
        d.work_date,
        d.agent_id,
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```
        d.hub_id,
        d.salary_type,
        d.active_status,
        COALESCE(e.assigned_attempts, 0) AS assigned_attempts,
        COALESCE(e.delivered_attempts, 0) AS delivered_attempts,
        COALESCE(e.hold_attempts, 0) AS hold_attempts,
        COALESCE(e.returned_attempts, 0) AS returned_attempts
    FROM agent_days d
    LEFT JOIN daily_events e
      ON e.agent_id = d.agent_id
     AND e.hub_id = d.hub_id
     AND e.work_date = d.work_date
)
SELECT *
FROM q1_daily
ORDER BY work_date, hub_id, agent_id
LIMIT 20;

-- Q1 full hub summary: deliveries per available agent-day.
WITH dates AS (
    SELECT d::date AS work_date
    FROM generate_series(DATE '2026-10-13', DATE '2026-10-26', INTERVAL '1 day') AS s(d)
),
agent_days AS (
    SELECT a.agent_id, a.hub_id, d.work_date
    FROM agents a
    CROSS JOIN dates d
    WHERE a.active_status = 'Active'
       OR (a.active_status = 'Inactive' AND d.work_date <= DATE '2026-10-19')
),
daily_events AS (
    SELECT
        o.hub_id,
        l.agent_id,
        l.event_time::date AS work_date,
        COUNT(*) FILTER (WHERE l.status = 'Delivered') AS delivered_attempts
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    WHERE l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY o.hub_id, l.agent_id, l.event_time::date
)
SELECT
    h.hub_id,
    h.hub_name,
    h.cluster,
    SUM(COALESCE(e.delivered_attempts, 0)) AS delivered_attempts,
    COUNT(*) AS available_agent_days,
    ROUND(
        SUM(COALESCE(e.delivered_attempts, 0))::numeric / NULLIF(COUNT(*), 0),
        2
    ) AS avg_deliveries_per_agent_per_day
FROM agent_days a
JOIN hubs h ON h.hub_id = a.hub_id
LEFT JOIN daily_events e
  ON e.agent_id = a.agent_id
 AND e.hub_id = a.hub_id
 AND e.work_date = a.work_date
GROUP BY h.hub_id, h.hub_name, h.cluster
ORDER BY h.hub_id;
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```sql
-- Q2. Hub utilisation in Week 1 and Week 2
-- Week 1 includes inactive agents; Week 2 excludes them.
-- ============================================================

WITH headcount AS (
    SELECT
        hub_id,
        COUNT(*) AS week1_agents,
        COUNT(*) FILTER (WHERE active_status = 'Active') AS week2_agents
    FROM agents
    GROUP BY hub_id
),
attempts AS (
    SELECT
        o.hub_id,
        COUNT(*) FILTER (
            WHERE l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-19'
        ) AS week1_attempts,
        COUNT(*) FILTER (
            WHERE l.event_time::date BETWEEN DATE '2026-10-20' AND DATE '2026-10-26'
        ) AS week2_attempts
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    WHERE l.status = 'Assigned'
      AND l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY o.hub_id
)
SELECT
    h.hub_id,
    h.hub_name,
    h.cluster,
    a.week1_attempts,
    a.week2_attempts,
    c.week1_agents,
    c.week2_agents,
    ROUND(a.week1_attempts::numeric / (c.week1_agents * 40 * 7), 4) AS week1_utilisation,
    ROUND(a.week2_attempts::numeric / (c.week2_agents * 40 * 7), 4) AS week2_utilisation,
    ROUND((a.week2_attempts::numeric / (c.week2_agents * 40 * 7)) * 100, 2) > 85
        AS week2_above_85pct_target,
    ROUND((a.week2_attempts::numeric / NULLIF(a.week1_attempts, 0) - 1) * 100, 2)
        AS attempt_change_pct
FROM hubs h
JOIN headcount c ON c.hub_id = h.hub_id
JOIN attempts a ON a.hub_id = h.hub_id
ORDER BY h.hub_id;

-- ============================================================
-- Q3. Week 2 Hold Reasons
-- Agent-side reasons per case notes: Late Dispatch, Not Attempted,
-- Agent Time Shortage. All others are customer-side.
-- ============================================================

WITH week2_holds AS (
    SELECT
        o.hub_id,
        l.hold_reason,
        CASE
            WHEN l.hold_reason IN ('Late Dispatch', 'Not Attempted', 'Agent Time Shortage')
                THEN 'Agent-side'
            ELSE 'Customer-side'
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    WHERE l.status = 'Hold'
      AND l.event_time::date BETWEEN DATE '2026-10-20' AND DATE '2026-10-26'
),
hub_totals AS (
    SELECT
        hub_id,
        COUNT(*) AS total_holds,
        COUNT(*) FILTER (WHERE hold_side = 'Agent-side') AS agent_side_holds
    FROM week2_holds
    GROUP BY hub_id
),
reason_counts AS (
    SELECT hub_id, hold_reason, COUNT(*) AS hold_count
    FROM week2_holds
    GROUP BY hub_id, hold_reason
),
ranked_reasons AS (
    SELECT *,
        ROW_NUMBER() OVER (
            PARTITION BY hub_id
            ORDER BY hold_count DESC, hold_reason
        ) AS reason_rank
    FROM reason_counts
),
top_two AS (
    SELECT
        hub_id,
        MAX(hold_reason) FILTER (WHERE reason_rank = 1) AS top_1_reason,
        MAX(hold_count) FILTER (WHERE reason_rank = 1) AS top_1_count,
        MAX(hold_reason) FILTER (WHERE reason_rank = 2) AS top_2_reason,
        MAX(hold_count) FILTER (WHERE reason_rank = 2) AS top_2_count
    FROM ranked_reasons
    WHERE reason_rank <= 2
    GROUP BY hub_id
)
SELECT
    h.hub_id,
    h.hub_name,
    t.total_holds,
    t.agent_side_holds,
    ROUND(t.agent_side_holds::numeric / NULLIF(t.total_holds, 0), 4) AS agent_side_share,
    r.top_1_reason,
    r.top_1_count,
    r.top_2_reason,
    r.top_2_count
FROM hub_totals t
JOIN hubs h ON h.hub_id = t.hub_id
JOIN top_two r ON r.hub_id = t.hub_id
ORDER BY h.hub_id;

-- ============================================================
-- Q4. What Happens to Held Orders
-- Count unique orders held at least once; classify by orders.final_status.
-- ============================================================

WITH held_order_ids AS (
    SELECT DISTINCT order_id
    FROM order_logs
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```
),
held_orders AS (
    SELECT o.*
    FROM orders o
    JOIN held_order_ids h ON h.order_id = o.order_id
)
SELECT
    h.hub_id,
    h.hub_name,
    COUNT(*) AS held_orders,
    COUNT(*) FILTER (WHERE o.final_status = 'Delivered') AS finally_delivered,
    ROUND(
        COUNT(*) FILTER (WHERE o.final_status = 'Delivered')::numeric / COUNT(*), 4
    ) AS delivered_share,
    COUNT(*) FILTER (WHERE o.final_status = 'Returned') AS finally_returned,
    ROUND(
        COUNT(*) FILTER (WHERE o.final_status = 'Returned')::numeric / COUNT(*), 4
    ) AS returned_share,
    COUNT(*) FILTER (WHERE o.final_status = 'Hold') AS still_on_hold,
    ROUND(
        COUNT(*) FILTER (WHERE o.final_status = 'Hold')::numeric / COUNT(*), 4
    ) AS still_on_hold_share
FROM held_orders o
JOIN hubs h ON h.hub_id = o.hub_id
GROUP BY h.hub_id, h.hub_name
ORDER BY h.hub_id;

-- ============================================================
-- Q5. High-Return Agents
-- Return ratio = Returned outcome attempts / Assigned attempts.
-- Tenure must be strictly greater than 30 days on 26 October 2026.
-- ============================================================

WITH agent_counts AS (
    SELECT
        agent_id,
        COUNT(*) FILTER (WHERE status = 'Assigned') AS assigned_attempts,
        COUNT(*) FILTER (WHERE status = 'Returned') AS returned_attempts
    FROM order_logs
    WHERE event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY agent_id
)
SELECT
    a.agent_id,
    a.agent_name,
    a.hub_id,
    a.salary_type,
    DATE '2026-10-26' - a.joining_date AS tenure_days,
    c.assigned_attempts,
    c.returned_attempts,
    ROUND(c.returned_attempts::numeric / NULLIF(c.assigned_attempts, 0), 4) AS return_ratio
FROM agents a
JOIN agent_counts c ON c.agent_id = a.agent_id
WHERE DATE '2026-10-26' - a.joining_date > 30
  AND c.returned_attempts::numeric / NULLIF(c.assigned_attempts, 0) > 0.15
ORDER BY return_ratio DESC, c.returned_attempts DESC;

-- ============================================================
-- Q6. Cost per Delivered Order
-- Workday = agent-day with >=1 Assigned event. A Delivered outcome
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```sql
-- Q4: Net delivery profit per hub per week
-- Excludes COD and delivery_charge because they are not operating
-- costs in the company cost sheet.
-- ============================================================

WITH rule_values AS (
    SELECT
        MAX(value::numeric) FILTER (WHERE rule = 'Working days per month') AS working_days_per_month
    FROM rules
),
salary AS (
    SELECT
        MAX(monthly_salary) FILTER (WHERE salary_type = 'Fixed') AS fixed_monthly_salary,
        MAX(daily_allowance) FILTER (WHERE salary_type = 'Variable') AS variable_daily_allowance,
        MAX(allowance_per_delivery) FILTER (WHERE salary_type = 'Variable') AS variable_per_delivery
    FROM agent_salary
),
cost_values AS (
    SELECT
        MAX(amount_bdt) FILTER (WHERE what = 'A hold') AS hold_cost,
        MAX(amount_bdt) FILTER (WHERE what = 'A return') AS return_cost
    FROM costs
),
assigned_days AS (
    SELECT DISTINCT
        o.hub_id,
        l.agent_id,
        l.event_time::date AS work_date,
        CASE WHEN l.event_time::date <= DATE '2026-10-19' THEN 'Week 1' ELSE 'Week 2' END AS week_name,
        a.salary_type
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    JOIN agents a ON a.agent_id = l.agent_id
    WHERE l.status = 'Assigned'
      AND l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
),
payroll AS (
    SELECT
        d.hub_id,
        d.week_name,
        SUM(
            CASE WHEN d.salary_type = 'Fixed'
                 THEN s.fixed_monthly_salary / r.working_days_per_month ELSE 0 END
        ) AS fixed_pay,
        SUM(
            CASE WHEN d.salary_type = 'Variable'
                 THEN s.variable_daily_allowance ELSE 0 END
        ) AS variable_daily_pay
    FROM assigned_days d
    CROSS JOIN salary s
    CROSS JOIN rule_values r
    GROUP BY d.hub_id, d.week_name
),
variable_deliveries AS (
    SELECT
        o.hub_id,
        CASE WHEN l.event_time::date <= DATE '2026-10-19' THEN 'Week 1' ELSE 'Week 2' END AS week_name,
        COUNT(*) AS variable_delivered_events
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    JOIN agents a ON a.agent_id = l.agent_id
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```
      AND a.salary_type = 'variable'
      AND l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY o.hub_id, week_name
),
operating_events AS (
    SELECT
        o.hub_id,
        CASE WHEN l.event_time::date <= DATE '2026-10-19' THEN 'Week 1' ELSE 'Week 2' END AS week_name,
        COUNT(*) FILTER (WHERE l.status = 'Delivered') AS delivered_orders,
        COUNT(*) FILTER (WHERE l.status = 'Hold') AS holds,
        COUNT(*) FILTER (WHERE l.status = 'Returned') AS returns
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
    WHERE l.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY o.hub_id, week_name
)
SELECT
    h.hub_id,
    h.hub_name,
    e.week_name,
    COALESCE(p.fixed_pay, 0) AS fixed_pay,
    COALESCE(p.variable_daily_pay, 0) AS variable_daily_pay,
    COALESCE(v.variable_delivered_events, 0) * s.variable_per_delivery AS variable_delivery_allowance,
    e.holds * c.hold_cost AS hold_cost,
    e.returns * c.return_cost AS return_cost,
    COALESCE(p.fixed_pay, 0)
      + COALESCE(p.variable_daily_pay, 0)
      + COALESCE(v.variable_delivered_events, 0) * s.variable_per_delivery
      + e.holds * c.hold_cost
      + e.returns * c.return_cost AS total_operating_cost,
    e.delivered_orders,
    ROUND(
        (
          COALESCE(p.fixed_pay, 0)
          + COALESCE(p.variable_daily_pay, 0)
          + COALESCE(v.variable_delivered_events, 0) * s.variable_per_delivery
          + e.holds * c.hold_cost
          + e.returns * c.return_cost
        ) / NULLIF(e.delivered_orders, 0),
        2
    ) AS cost_per_delivered_order
FROM operating_events e
JOIN hubs h ON h.hub_id = e.hub_id
LEFT JOIN payroll p ON p.hub_id = e.hub_id AND p.week_name = e.week_name
LEFT JOIN variable_deliveries v ON v.hub_id = e.hub_id AND v.week_name = e.week_name
CROSS JOIN salary s
CROSS JOIN cost_values c
ORDER BY h.hub_id, e.week_name;

-- ============================================================
-- Q7. Agents Needed for the Forecast
-- Forecast expected orders are converted to attempts using each
-- hub's observed 14-day Assigned attempts / unique original orders.
-- Required agents = CEIL(forecast attempts / (7 * 40 * 85%)).
-- ============================================================

WITH historical_attempts AS (
    SELECT o.hub_id, COUNT(*) AS assigned_attempts
    FROM order_logs l
    JOIN orders o ON o.order_id = l.order_id
```

APPENDIX A | EDITABLE POSTGRESQL SQL | QUERIES Q1-Q7

```
      AND t.event_time::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY o.hub_id
),
historical_orders AS (
    SELECT hub_id, COUNT(*) AS total_orders
    FROM orders
    WHERE created_at::date BETWEEN DATE '2026-10-13' AND DATE '2026-10-26'
    GROUP BY hub_id
),
attempt_rate AS (
    SELECT
        a.hub_id,
        a.assigned_attempts::numeric / NULLIF(o.total_orders, 0) AS attempts_per_order
    FROM historical_attempts a
    JOIN historical_orders o ON o.hub_id = a.hub_id
),
active_agents AS (
    SELECT hub_id, COUNT(*) AS active_agents
    FROM agents
    WHERE active_status = 'Active'
    GROUP BY hub_id
),
standard AS (
    SELECT
        MAX(value::numeric) FILTER (WHERE rule = 'Standard attempts per agent per day') AS attempts_per_agent_day,
        MAX(value::numeric) FILTER (WHERE rule = 'Target utilisation') AS target_utilisation
    FROM rules
)
SELECT
    h.hub_id,
    h.hub_name,
    h.cluster,
    f.week_start,
    f.week_end,
    f.expected_orders,
    ROUND(r.attempts_per_order, 4) AS attempts_per_order,
    ROUND(f.expected_orders * r.attempts_per_order, 1) AS forecast_attempts,
    CEIL(
        (f.expected_orders * r.attempts_per_order)
        / (7 * s.attempts_per_agent_day * s.target_utilisation)
    )::integer AS required_agents,
    a.active_agents,
    CEIL(
        (f.expected_orders * r.attempts_per_order)
        / (7 * s.attempts_per_agent_day * s.target_utilisation)
    )::integer - a.active_agents AS staffing_gap
FROM volume_forecast f
JOIN hubs h ON h.hub_id = f.hub_id
JOIN attempt_rate r ON r.hub_id = f.hub_id
JOIN active_agents a ON a.hub_id = f.hub_id
CROSS JOIN standard s
ORDER BY f.week_start, h.hub_id;
```